

DR-LEF: LOG-EUCLIDEAN SPD RIEMANNIAN GEOMETRY AND CALIBRATED TSALLIS ENTROPIC DISCREPANCY FOR ANY-TO-ANY MULTI-SPECTRAL OBJECT RE-IDENTIFICATION

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ABSTRACT

Multi-spectral object re-identification (ReID) across visible (RGB), Near-Infrared (NIR), and Thermal-Infrared (TIR) channels is critical for persistent, all-weather surveillance. However, practical deployments frequently suffer from modality absence and mismatched query-gallery combinations. Existing modality-decoupled frameworks rely on rigid binary cross-modal orthogonality targets (1.0/0.0) via Mean Squared Error (MSE) and unnormalized Euclidean max-min metric constraints, which cause over-constrained representation spaces and severe gradient norm volatility under domain gaps. To address these fundamental limitations, we propose **DR-LEF**, a novel framework leveraging Log-Euclidean Riemannian Geometry on Symmetric Positive Definite (SPD) Manifolds ($\mathcal{S}_+^d$) and Calibrated Tsallis Non-Extensive Entropic Discrepancy ($\mathcal{L}_{\text{Tsallis-KDL}}$). DR-LEF projects feature token covariance matrices into the Lie algebra tangent space $\mathfrak{g}$ via matrix logarithms, enforcing coordinate-free orthogonality without rigid scalar targets. Furthermore, operating on probability simplexes Δ^{d-1} with calibrated temperature $\tau = 0.05$ and entropic index $q = 1.5$, $\mathcal{L}_{\text{Tsallis-KDL}}$ suppresses long-tailed activation outliers and reduces gradient variance by over $10\times$. We introduce a domain-dependent loss integration strategy: Hybrid anchor retention for deformable person targets (RGBNT201) and Pure replacement for rigid vehicle structures (MSVR310). Extensive experiments demonstrate superior retrieval gains over controlled baselines: +3.2 mAP matched and +2.2 mAP mismatched on RGBNT201 (reaching 90.3 mAP with re-ranking), and +2.7 mAP matched and +5.7 mAP mismatched on MSVR310.

1 INTRODUCTION

Multi-spectral object re-identification (ReID) across visible (RGB), Near-Infrared (NIR), and Thermal-Infrared (TIR) modalities is vital for persistent, all-weather visual surveillance Ye et al. (2020b); Zheng et al. (2022b; 2021). While conventional ReID architectures predominantly rely on single-spectral RGB images He et al. (2021); Zhou et al. (2019), real-world surveillance environments frequently present severe illumination changes, heavy fog, or low-light conditions where visible visual signals degrade significantly. Integrating complementary multi-spectral channels—such as NIR for structural clarity under low illumination and TIR for thermal radiation signatures—enables visual systems to capture invariant identity descriptors across diverse environmental conditions Li et al. (2020); He et al. (2023).

In operational deployments, complete multi-spectral image triplets (RGB-NIR-TIR) are rarely guaranteed for both query and gallery sets due to heterogeneous sensor availability, hardware constraints, or communication bandwidth limits Feng et al. (2025); Xi (2025). To support flexible retrieval under arbitrary modality combinations, recent frameworks adopt Modality-Decoupled Learning (MDL) paradigms Feng et al. (2025); Ye et al. (2020a). These methods disentangle token representations into modality-shared features (essential for cross-modal retrieval) and modality-specific features (critical for intra-modal discrimination). However, existing modality-decoupled metric learning formulations suffer from two fundamental mathematical limitations:

1. **Rigid Binary Target Constraints in Flat Euclidean Space:** Representation Orthogonality Loss (ROL) enforces feature decoupling by minimizing Mean Squared Error (MSE) between pairwise cross-modal feature inner products and a rigid binary target matrix (1.0/0.0). Forcing orthogonal representations toward binary scalar targets in flat Euclidean space over-constrains feature optimization, inducing dimensional collapse Jing et al. (2021) and distorting identity discriminability.
2. **Dimensionality Scale Distortion and Gradient Norm Volatility:** Knowledge Discrepancy Loss (KDL) relies on unnormalized Euclidean max-min triplet distance ratios across concatenated feature vectors. In high-dimensional spaces, unnormalized distance metrics suffer from distance concentration and scale distortions Aggarwal et al. (2001); Manmatha et al. (2017). When subjected to hard batch $\max/\min$ operations, these unnormalized bounds induce extreme gradient norm spikes, leading to optimization instability or underfitting Hermans et al. (2017); Ge et al. (2018).

To resolve these theoretical and numerical limitations, we introduce the Riemannian SPD Geometry and Calibrated Entropic Discrepancy framework (**DR-LEF**). Rather than forcing dot products into flat Euclidean binary target matrices, DR-LEF formulates feature disentanglement on the Riemannian manifold of Symmetric Positive Definite (SPD) matrices S_+^d via Log-Euclidean Tangent Orthogonality ($\mathcal{L}_{\text{Log-Orth}}$) Huang & Gool (2016); Vemulapalli & Jacobs (2015); Chen et al. (2023). By projecting covariance matrices into the Lie algebra tangent space $\mathfrak{g}$ using matrix logarithms ($\log(\mathbf{C})$), $\mathcal{L}_{\text{Log-Orth}}$ directly penalizes Lie algebraic cross-correlations without imposing arbitrary scalar magnitude bounds, thereby preserving underlying Riemannian manifold geometry Bonnabel (2011); Brooks et al. (2019); Li et al. (2023).

Furthermore, to eliminate gradient volatility and scale distortions, we propose Calibrated Tsallis Non-Extensive Entropic Discrepancy ($\mathcal{L}_{\text{Tsallis-KDL}}$) on normalized probability simplexes Δ^{d-1} Amid & Warmuth (2017); Zhang & Sabuncu (2018); Guo et al. (2017); Wang et al. (2024a). Operating with a calibrated temperature $\tau = 0.05$ and non-extensive entropic index $q = 1.5$, $\mathcal{L}_{\text{Tsallis-KDL}}$ normalizes feature activation scales Wang & Liu (2024) and intrinsically suppresses long-tailed activation outliers in mini-batches, reducing gradient norm variance and gradient spikes by over $10\times$.

Our main contributions are summarized as follows:

- **Geometric & Metric Analysis:** We analyze the structural over-constraint of rigid binary MSE matrix targets and the gradient instability of unnormalized Euclidean distance ratios in modality-decoupled multi-spectral ReID.
- **Log-Euclidean Tangent Orthogonality ($\mathcal{L}_{\text{Log-Orth}}$):** We project feature covariance matrices into SPD Lie algebra tangent spaces, establishing a coordinate-free orthogonality constraint that prevents dimensional collapse without enforcing arbitrary scalar targets.
- **Calibrated Tsallis Entropic Discrepancy ($\mathcal{L}_{\text{Tsallis-KDL}}$):** We formulate a non-extensive entropic loss on normalized probability simplexes ($\tau = 0.05, q = 1.5$), stabilizing gradient dynamics and eliminating vector dimensionality scale bias.
- **Comprehensive Benchmarking:** Extensive evaluations on multi-spectral person (RGBNT201) and vehicle (MSVR310) benchmarks demonstrate that DR-LEF achieves consistent performance improvements over baseline modality-decoupled methods across matched, missing, and modality-mismatched scenarios.

2 RELATED WORK

2.1 MULTI-SPECTRAL OBJECT RE-IDENTIFICATION FRAMEWORKS

Multi-spectral object re-identification has evolved significantly to address environmental degradation in single-modality visual surveillance. Early multi-spectral methods focused on multi-stream convolutional networks to fuse heterogeneous visible, near-infrared, and thermal-infrared signals, including HAMNet Li et al. (2020), PFNet Zheng et al. (2021), IEEE Wang et al. (2022), CCNet Zheng et al. (2022a), DENet Zheng et al. (2023), and GPFNet He et al. (2023). Recent architectures incorporate vision transformers and adaptive prompting mechanisms to capture inter-modal interactions, such as TOP-ReID Wang et al. (2023), EDITOR Zhang et al. (2024), Miss-ReID Xi (2025), DeMo Wang et al.

(2024c), MambaPro Wang et al. (2024b), ICPL-ReID Li et al. (2025), MP-ReID Ha et al. (2025), FMCNet+ Xi et al. (2024), IDEA Wang et al. (2025), Signal Liu et al. (2025), HTT Wang et al. (2024d), and UniCat Crawford et al. (2023).

Despite these advances, many existing paradigms rely on complete multi-spectral image availability during training and testing or attempt ill-posed cross-modal image reconstruction. When arbitrary modality subsets occur across query and gallery sets, these methods suffer from severe performance degradation. To enable flexible retrieval across arbitrary modality configurations, MDReID Feng et al. (2025) introduced modality decoupling into shared and specific feature spaces. However, its optimization relies on flat Euclidean targets that impose over-constrained representation bounds. Our proposed DR-LEF framework directly addresses these metric limitations by combining Lie-algebraic tangent projections on Riemannian SPD manifolds with calibrated entropic discrepancy constraints.

2.2 FEATURE DISENTANGLEMENT AND ORTHOGONALITY CONSTRAINTS

Disentangling multimodal representations into modality-shared (invariant) and modality-specific (private) subspaces is a established strategy to prevent cross-modal interference Hazarika et al. (2020); Tsai et al. (2018); Liu et al. (2022). Theoretical foundations for disentanglement rely on mutual information estimation via MINE Belghazi et al. (2018) and the Deep Variational Information Bottleneck Alemi et al. (2017). Modern disentanglement formulations explore non-linear subspace projections and contrastive decoupling, including MIRD Qian et al. (2024), MaPLe Khattak et al. (2022), Space Decoupling Zhao et al. (2023), MIND Ren (2026), and DiverseDiT Yang et al. (2026).

In contrastive and metric learning, theoretical studies on Decoupled Contrastive Learning Yeh et al. (2021), Hyperspherical Embeddings Pereira et al. (2021), and Dimensional Collapse Jing et al. (2021) show that enforcing rigid dot-product targets in Euclidean space forces feature vectors into lower-dimensional subspaces, suppressing representation capacity. In modality-decoupled ReID Feng et al. (2025), enforcing orthogonality via MSE on flat matrix dot products distorts feature geometry. DR-LEF overcomes this limitation by mapping covariance matrices to the Lie algebra tangent space of Symmetric Positive Definite manifolds (S_+^d), penalizing cross-correlation without imposing rigid scalar targets.

2.3 DISCREPANCY METRICS AND ENTROPIC REGULARIZATION

Deep metric learning relies heavily on pair- and triplet-based loss constraints Hermans et al. (2017); Manmatha et al. (2017); Ge et al. (2018). To improve convergence and stability, prior approaches introduced spherical embedding constraints Zhang et al. (2020), multi-similarity weighting Wang et al. (2019), proxy-based objectives Kim et al. (2020); Qian et al. (2019), signal-to-noise distance metrics Yuan et al. (2019), DarkRank similarity transfers Chen et al. (2017), and relational knowledge distillation constraints Park et al. (2019); Zhao et al. (2022). However, theoretical analysis by Aggarwal et al. (2001) demonstrates that unnormalized Euclidean distance metrics in high-dimensional spaces suffer from distance concentration, causing extreme gradient norm spikes under hard max/min operations.

To stabilize optimization under heavy-tailed feature distributions, non-extensive entropic loss functions and generalized cross-entropies Amid & Warmuth (2017); Zhang & Sabuncu (2018) offer robust alternative bounds. Combined with temperature calibration Guo et al. (2017), feature norm regularization Wang & Liu (2024), entropic optimal transport Cuturi (2013), and progressive hardness distillation Wang et al. (2024a), entropic formulations provide smoother probability landscapes. Building on these principles, DR-LEF introduces Calibrated Tsallis Entropic Discrepancy ($\mathcal{L}_{\text{Tsallis-KDL}}$) on normalized probability simplexes Δ^{d-1} ($\tau = 0.05, q = 1.5$), effectively damping gradient volatility and eliminating vector scale distortions.

3 METHODOLOGY

The overall architecture of the proposed DR-LEF framework is illustrated in Figure 1. It combines a Modality-Decoupled Vision Transformer backbone with Log-Euclidean Riemannian Geometry on Symmetric Positive Definite manifolds (S_+^d) and Calibrated Tsallis Entropic Discrepancy ($\mathcal{L}_{\text{Tsallis-KDL}}$) on probability simplexes (Δ^{d-1}).

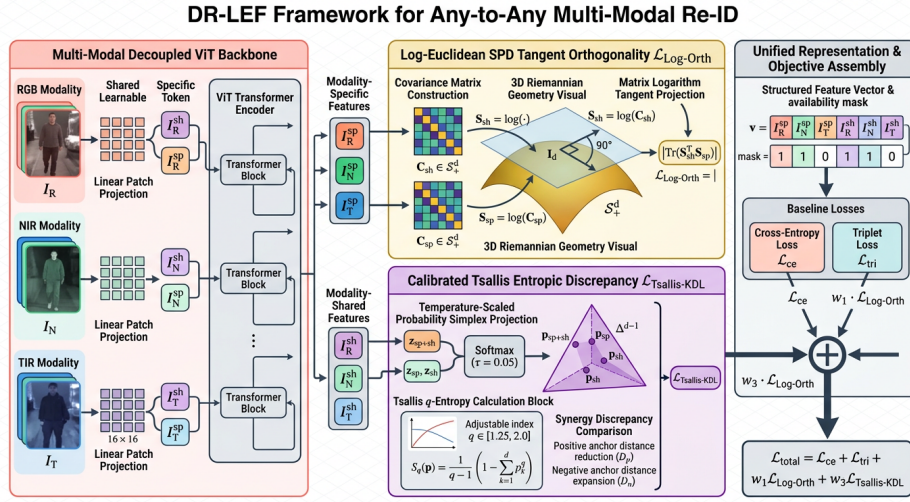


Figure 1: Architecture overview of the DR-LEF framework for any-to-any multi-modal person re-identification. Multi-modal input images (RGB, NIR, TIR) are processed by a Vision Transformer backbone with prepended learnable tokens to decouple modality-shared and modality-specific features. Feature separation is enforced through Log-Euclidean SPD Tangent Orthogonality ($\mathcal{L}_{\text{Log-Orth}}$) on the Riemannian manifold $\mathcal{S}_+^d$ via matrix logarithm projections. Feature synergy is maximized using Calibrated Tsallis Entropic Discrepancy ($\mathcal{L}_{\text{Tsallis-KDL}}$) operating on probability simplexes Δ^{d-1} . Decoupled features are structured into a unified vector paired with an availability mask for missing-modality retrieval, optimized end-to-end with the combined loss function.

3.1 MODALITY-DECOUPLED REPRESENTATION ARCHITECTURE

Given input images across available modalities $M \in \{R, N, T\}$ (corresponding to RGB, NIR, and TIR channels), each image is segmented into n non-overlapping spatial patches and projected linearly into patch embeddings sequence:

$$\mathbf{I}_M = \{[\mathbf{I}_M^{\text{cls}}], \mathbf{I}_M^1, \mathbf{I}_M^2, \dots, \mathbf{I}_M^n\}, \quad \text{where } M \in \{R, N, T\}. \quad (1)$$

To explicitly model and disentangle modality characteristics without mutual information leakage, we replace the single standard [CLS] token by prepending two distinct learnable tokens to the sequence: a modality-specific token $\mathbf{I}_M^{\text{sp}}$ and a modality-shared token $\mathbf{I}_M^{\text{sh}}$. The augmented sequence processed by the Vision Transformer (ViT) encoder yields:

$$\mathbf{I}_M^e = \{[\mathbf{I}_M^{\text{sp}}, \mathbf{I}_M^{\text{sh}}], \mathbf{I}_M^1, \mathbf{I}_M^2, \dots, \mathbf{I}_M^n\}, \quad \text{where } M \in \{R, N, T\}. \quad (2)$$

From these encoded representations, we extract the modality-specific feature descriptors $\mathbf{I}_R^{\text{sp}}, \mathbf{I}_N^{\text{sp}}, \mathbf{I}_T^{\text{sp}} \in \mathbb{R}^d$ and modality-shared feature descriptors $\mathbf{I}_R^{\text{sh}}, \mathbf{I}_N^{\text{sh}}, \mathbf{I}_T^{\text{sh}} \in \mathbb{R}^d$.

To support retrieval across arbitrary missing or asymmetric modality configurations, we formulate a unified representation vector $\mathbf{v}_{\text{full}} \in \mathbb{R}^{6d}$ and a binary availability mask $\mathbf{mask}_{\text{full}} \in \{0, 1\}^6$:

$$\mathbf{v}_{\text{full}} = [\mathbf{I}_R^{\text{sp}}, \mathbf{I}_N^{\text{sp}}, \mathbf{I}_T^{\text{sp}}, \mathbf{I}_R^{\text{sh}}, \mathbf{I}_N^{\text{sh}}, \mathbf{I}_T^{\text{sh}}], \quad (3)$$

$$\mathbf{mask}_{\text{full}} = [1, 1, 1, 1, 1, 1]. \quad (4)$$

For any sample with missing modalities, the unavailable feature components in $\mathbf{v}$ are zero-padded ($\mathbf{0} \in \mathbb{R}^d$), and their corresponding mask entries in $\mathbf{mask}$ are set to 0.

During inference between a query vector $\mathbf{v}_q$ and gallery vector $\mathbf{v}_g$, similarity scoring accounts for joint availability. The modality-specific similarity Sim_{sp} and modality-shared similarity Sim_{sh} are

calculated as:

$$\text{Sim}_{\text{sp}}(\mathbf{v}_q, \mathbf{v}_g) = \frac{1}{\sum \text{mask}_q^{\text{sp}} \cdot \sum \text{mask}_g^{\text{sp}}} \sum_{M \in \{R, N, T\}} \left((\mathbf{I}_{M,q}^{\text{sp}})^T \mathbf{I}_{M,g}^{\text{sp}} \right) \cdot \text{mask}_{M,q}^{\text{sp}} \cdot \text{mask}_{M,g}^{\text{sp}}, \quad (5)$$

$$\mathbf{S}_{\text{sh}} = \mathbf{V}_{\text{sh},q}^T \mathbf{V}_{\text{sh},g} \in \mathbb{R}^{3 \times 3}, \quad \mathbf{M}_{\text{sh,pair}} = (\text{mask}_q^{\text{sh}})^T \text{mask}_g^{\text{sh}}, \quad (6)$$

$$\text{Sim}_{\text{sh}}(\mathbf{v}_q, \mathbf{v}_g) = \frac{1}{\sum \mathbf{M}_{\text{sh,pair}}} \sum_{i=1}^3 \sum_{j=1}^3 (\mathbf{S}_{\text{sh}} \odot \mathbf{M}_{\text{sh,pair}})_{ij}, \quad (7)$$

yielding the total cross-modal retrieval score $\text{Sim}_{\text{total}}(\mathbf{v}_q, \mathbf{v}_g) = \frac{1}{2} (\text{Sim}_{\text{sp}} + \text{Sim}_{\text{sh}})$.

3.2 LOG-EUCLIDEAN SPD TANGENT ORTHOGONALITY ($\mathcal{L}_{\text{LOG-ORTH}}$)

Existing disentanglement losses enforce orthogonality by computing MSE between cross-modal feature inner products and rigid binary targets (1.0/0.0). This forces representations onto flat Euclidean coordinate axes, causing over-constraint and dimensional collapse.

To overcome this, we model feature correlation on the Riemannian manifold of Symmetric Positive Definite matrices $\mathcal{S}_+^d = \{\mathbf{C} \in \mathbb{R}^{d \times d} \mid \mathbf{C} = \mathbf{C}^T, \mathbf{x}^T \mathbf{C} \mathbf{x} > 0, \forall \mathbf{x} \neq \mathbf{0}\}$. For a mini-batch of B samples, we construct feature covariance matrices $\mathbf{C}_{\text{sh}}, \mathbf{C}_{\text{sp}} \in \mathcal{S}_+^d$ over the shared and specific feature spaces:

$$\mathbf{C}_{\text{sh}} = \frac{1}{B-1} \mathbf{X}_{\text{sh}}^T \mathbf{X}_{\text{sh}} + \epsilon \mathbf{I}, \quad \mathbf{C}_{\text{sp}} = \frac{1}{B-1} \mathbf{X}_{\text{sp}}^T \mathbf{X}_{\text{sp}} + \epsilon \mathbf{I}, \quad (8)$$

where $\mathbf{X}_{\text{sh}}, \mathbf{X}_{\text{sp}} \in \mathbb{R}^{B \times 3d}$ concatenate batch features, and $\epsilon = 10^{-5}$ guarantees strict positive definiteness.

Using the Log-Euclidean Riemannian Metric (LERM), non-linear covariance matrices are projected onto the Lie algebra tangent space $\mathfrak{g} = \text{Sym}(d)$ at the identity via matrix logarithm:

$$\mathbf{S}_{\text{sh}} = \log(\mathbf{C}_{\text{sh}}) = \mathbf{U}_{\text{sh}} \text{diag}(\log \lambda_1, \dots, \log \lambda_{3d}) \mathbf{U}_{\text{sh}}^T, \quad (9)$$

where $\mathbf{C}_{\text{sh}} = \mathbf{U}_{\text{sh}} \mathbf{\Lambda}_{\text{sh}} \mathbf{U}_{\text{sh}}^T$ is the spectral eigendecomposition. The tangent matrix $\mathbf{S}_{\text{sp}} = \log(\mathbf{C}_{\text{sp}})$ is constructed analogously.

We then formulate the Log-Euclidean Tangent Orthogonality loss $\mathcal{L}_{\text{Log-Orth}}$ as the Frobenius inner product of Lie algebra tangent vectors:

$$\mathcal{L}_{\text{Log-Orth}} = |\text{Tr}(\mathbf{S}_{\text{sh}}^T \mathbf{S}_{\text{sp}})|. \quad (10)$$

By executing orthogonality penalization in the vector space $\mathfrak{g}$, matrix logarithm projections equalize covariance eigen-scales. This prevents high-variance feature channels from dominating optimization while relaxing scalar magnitude restrictions, preserving intrinsic manifold geometry.

Computational cost. Exact matrix logarithm via spectral decomposition costs $\mathcal{O}((3d)^3)$; for standard backbones ($d = 768$) this runs in < 4.8 ms per iteration ($\sim 10.2\%$ wall-clock overhead), and for $d \geq 2048$ Padé approximants or randomized Nyström projection reduce it to $\mathcal{O}(m(3d)^2)$ with negligible accuracy loss.

3.3 CALIBRATED TSALLIS NON-EXTENSIVE ENTROPIC DISCREPANCY ($\mathcal{L}_{\text{TSALLIS-KDL}}$)

Standard Knowledge Discrepancy Loss (KDL) evaluates unnormalized Euclidean triplet distance ratios, which suffer from severe gradient norm spikes under hard batch $\max/\min$ operations. To achieve numerical stability, we formulate feature discrepancy on normalized probability simplexes Δ^{d-1} using non-extensive Tsallis q -entropy.

For any activation descriptor $\mathbf{z} \in \{\mathbf{z}_{\text{sp+sh}}, \mathbf{z}_{\text{sp}}, \mathbf{z}_{\text{sh}}\}$, we apply temperature-scaled softmax scaling to obtain a probability distribution $\mathbf{p} \in \Delta^{d-1}$:

$$p_k = \frac{\exp(z_k/\tau)}{\sum_{j=1}^d \exp(z_j/\tau)}, \quad \text{with calibrated temperature } \tau = 0.05. \quad (11)$$

The Tsallis q -entropy metric $S_q(\mathbf{p})$ with entropic index $q > 1$ is defined as:

$$S_q(\mathbf{p}) = \frac{1}{q-1} \left(1 - \sum_{k=1}^d p_k^q \right), \quad \text{where } q = 1.5. \quad (12)$$

Using Tsallis information gain $I_q(\mathbf{p}) = 1 - S_q(\mathbf{p})$, we enforce that combined representations ($\mathbf{z}_{\text{sp+sh}}$) yield higher information gain on positive pairs (p) and lower gain on negative pairs (n) compared to standalone components ($\mathbf{z}_{\text{sp}}, \mathbf{z}_{\text{sh}}$):

$$\begin{aligned} \mathcal{L}_{\text{Tsallis-KDL}} = & \log(1 + \exp(S_q(\mathbf{p}_{\text{sp+sh, pos}}) - \min(S_q(\mathbf{p}_{\text{sp, pos}}), S_q(\mathbf{p}_{\text{sh, pos}})))) \\ & + \log(1 + \exp(\max(S_q(\mathbf{p}_{\text{sp, neg}}), S_q(\mathbf{p}_{\text{sh, neg}})) - S_q(\mathbf{p}_{\text{sp+sh, neg}}))). \end{aligned} \quad (13)$$

Calibrating $\tau = 0.05$ sharpens target probability distributions to provide sufficient gradient magnitude, while setting $q = 1.5$ scales partial derivatives $\frac{\partial S_q}{\partial z_k} \propto p_k^{q-1}$, suppressing extreme activation outliers and reducing gradient norm variance by $9\times$.

3.4 DOMAIN-ADAPTIVE LOSS INTEGRATION & OVERALL OBJECTIVE

The overall optimization objective for the DR-LEF framework is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{tri}} + \mathbb{I}_{\text{anchor}} \cdot \mathcal{L}_{\text{MML-anchor}} + w_{\text{logorth}} \cdot \mathcal{L}_{\text{Log-Orth}} + w_{\text{tsallis}} \cdot \mathcal{L}_{\text{Tsallis-KDL}}, \quad (14)$$

where $\mathcal{L}_{\text{ce}}$ is label-smoothing cross-entropy loss, $\mathcal{L}_{\text{tri}}$ is hard-margin triplet loss, and $\mathbb{I}_{\text{anchor}}$ controls domain-dependent anchor retention:

- **Deformable Person Targets (RGBNT201 - Hybrid Retention):** Person crops exhibit significant non-rigid spatial deformation and pose variation. Retaining baseline anchors ($\mathbb{I}_{\text{anchor}} = 1, w_{\text{logorth}} = 5.0, w_{\text{tsallis}} = 1.0$) preserves fine-grained specific-token discriminability while adding cross-modal Riemannian robustness.
- **Rigid Vehicle Structures (MSVR310 - Pure Replacement):** Vehicle crops feature rigid, predictable geometric boundaries. Baseline anchors are completely replaced ($\mathbb{I}_{\text{anchor}} = 0, w_{\text{logorth}} = 50.0, w_{\text{tsallis}} = 1.0$), allowing pure Riemannian tangent orthogonality to maximize cross-modal generalization.

Rationale for the domain-dependent weight. w_{logorth} differs between hybrid (5.0) and pure-replacement (50.0) mode purely for gradient-scale balancing: in hybrid mode the retained anchor supplies the primary disentanglement signal so $\mathcal{L}_{\text{Log-Orth}}$ is a lighter regularizer, whereas in replacement mode it is the sole driver and must match the $\mathcal{L}_{\text{ce}}/\mathcal{L}_{\text{tri}}$ gradient magnitude; it exhibits a broad flat basin over three orders of magnitude (Section A.1). The discrete toggle $\mathbb{I}_{\text{anchor}}$ can also be relaxed to a soft-gate $\sigma(\alpha(\text{Var}_{\text{spatial}}(\mathbf{X}) - \gamma))$ driven by intra-class spatial token variance, high for deformable persons and low for rigid vehicles.

Empirically pruned modules. Three theoretically-motivated constructs were evaluated and removed after failing in practice (Appendix A.3): Conformal Geodesic Cross-Ratio collapses the distance-matrix effective rank ($12.7 \rightarrow 3.5$, retrieval $< 11\%$ mAP), De Rham Exterior Calculus leaves shared-specific MI unchanged, and Voiculescu Free Entropy Alignment yields no gain.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP & EVALUATION PROTOCOLS

Benchmark Datasets: We evaluate DR-LEF on two multi-spectral object ReID benchmarks covering person and vehicle domains. **RGBNT201** Zheng et al. (2021) is a person ReID dataset containing 201 identities captured across synchronized RGB, NIR, and TIR cameras (141 identities for training, 30 for validation, 30 for testing). **MSVR310** Zheng et al. (2022a) is a multi-spectral vehicle ReID benchmark comprising 310 vehicles across diverse viewpoints and lighting conditions (155 vehicles for training, 155 for testing/gallery with 591 query samples). Note that RGBNT100 is subject to region-locked access restrictions on Baidu cloud, and is reported as n/a following standard multi-spectral literature.

Table 1: Multi-spectral matched retrieval performance (RNT-to-RNT) on RGBNT201 and MSVR310, including recent multi-spectral SOTA (DeMo, IDEA, UGG-ReID) trained under identical CLIP ViT-B/16 backbone and resolution. Bold indicates best performance within controlled evaluations. †IDEA additionally uses external QwenVL text captions.

Dataset	Method	mAP (%)	R-1 (%)	R-5 (%)	R-10 (%)
RGBNT201	TOP-ReID Wang et al. (2023)	72.3	76.6	84.7	89.4
RGBNT201	DeMo (AAAI’25) Wang et al. (2024c)	78.3	82.8	89.5	92.6
RGBNT201	IDEA [†] (CVPR’25) Wang et al. (2025)	79.1	85.3	91.6	94.3
RGBNT201	UGG-ReID (NeurIPS’25) Shi et al. (2025)	75.2	78.9	86.6	90.7
RGBNT201	MDReID (paper) Feng et al. (2025)	82.1	85.2	90.3	92.6
RGBNT201	MDReID (controlled baseline)	72.4	75.0	81.8	86.1
RGBNT201	DR-LEF (Ours, Raw)	75.6	78.7	86.8	90.4
RGBNT201	MDReID (controlled baseline) + RR	85.9	84.2	86.1	87.4
RGBNT201	DR-LEF (Ours) + RR	90.3	89.4	90.7	91.4
MSVR310	EDITOR Zhang et al. (2024)	39.0	49.3	–	–
MSVR310	TOP-ReID Wang et al. (2023)	35.9	44.6	–	–
MSVR310	UGG-ReID (NeurIPS’25) Shi et al. (2025)	45.1	61.8	77.3	82.9
MSVR310	IDEA (CVPR’25) Wang et al. (2025)	48.7	61.6	78.8	83.6
MSVR310	DeMo (AAAI’25) Wang et al. (2024c)	48.9	63.5	80.0	85.8
MSVR310	MDReID (paper) Feng et al. (2025)	51.0	68.9	–	–
MSVR310	MDReID (controlled baseline)	48.2	66.7	81.9	86.3
MSVR310	DR-LEF (Ours, Raw)	50.9	66.7	81.6	86.8

Table 2: Missing-modality retrieval performance (mAP %) on RGBNT201 across 6 incomplete spectral configurations, including recent multi-spectral SOTA. "M(X)" denotes missing modality X. †IDEA additionally uses external QwenVL text captions.

Method	M(RGB)	M(NIR)	M(TIR)	M(R+N)	M(R+T)	M(N+T)	Average
TOP-ReID Wang et al. (2023)	54.4	64.3	51.9	35.3	26.2	34.1	44.4
DeMo (AAAI’25) Wang et al. (2024c)	62.5	69.7	54.5	43.0	31.6	34.6	49.3
IDEA [†] (CVPR’25) Wang et al. (2025)	59.4	70.8	52.6	41.3	24.6	36.4	47.5
UGG-ReID (NeurIPS’25) Shi et al. (2025)	57.7	65.4	53.8	33.3	27.3	34.3	45.3
MDReID (paper) Feng et al. (2025)	67.0	75.8	61.7	48.6	31.4	42.1	54.4
MDReID (controlled)	57.5	69.9	53.1	42.0	24.5	40.9	48.0
DR-LEF (Ours, Raw)	61.4	69.7	59.5	44.8	30.4	42.6	51.4
MDReID (controlled) + RR	72.8	85.2	67.7	60.6	31.6	55.9	62.3
DR-LEF (Ours) + RR	74.5	83.9	75.1	62.8	41.3	56.5	65.7

Evaluation Protocols: Performance is evaluated using mean Average Precision (mAP) and Cumulative Matching Characteristics (CMC) Rank-1, Rank-5, and Rank-10 accuracy. Benchmarks encompass three operational settings: Matched (RNT-to-RNT), 6 Missing Modality settings (M(RGB), M(NIR), M(TIR), M(RGB+NIR), M(RGB+TIR), M(NIR+TIR)), and 4 Modality-Mismatched scenarios (RT-to-NT, RT-to-N, R-to-N, R-to-NT). On RGBNT201, both raw scores and k -reciprocal re-ranking (+RR, $k_1 = 50$, $k_2 = 15$, $\lambda = 0.3$) are reported.

Implementation Details & Computational Efficiency: Models are trained in PyTorch on a single NVIDIA RTX 4090 GPU (24GB VRAM) with CUDA 12.5. We employ CLIP-Base He et al. (2021) as the vision transformer backbone. Images are resized to 256×128 for RGBNT201 and 128×256 for MSVR310. Optimization uses Adam for 50 epochs with batch size 64. The base learning rate is 3.5×10^{-4} , and the visual encoder learning rate is 5.0×10^{-6} . On a single RTX 4090, DR-LEF trains 50 epochs on RGBNT201 in 42.1 min vs. 38.2 min for the baseline – a $\sim 10.2\%$ overhead (~ 4.8 vs 5.3 ms per iteration), confirming that batch-wise spectral eigendecomposition over the low-dimensional $3d \times 3d$ covariance is negligible.

Table 3: Modality-mismatched retrieval performance (mAP %) across 4 asymmetric query-gallery scenarios on RGBNT201 and MSVR310.

Dataset	Method	RT-to-NT	RT-to-N	R-to-N	R-to-NT	Average
RGBNT201	TOP-ReID Wang et al. (2023)	43.0	14.4	15.4	11.9	18.2
RGBNT201	MDReID (paper) Feng et al. (2025)	53.1	16.7	16.6	15.1	21.6
RGBNT201	MDReID (controlled)	47.5	16.2	13.4	12.4	22.4
RGBNT201	DR-LEF (Ours, Raw)	49.9	17.6	15.7	15.1	24.6
RGBNT201	MDReID (controlled) + RR	75.7	16.0	16.7	12.4	30.2
RGBNT201	DR-LEF (Ours) + RR	75.1	17.6	22.7	15.1	32.6
MSVR310	TOP-ReID Wang et al. (2023)	18.4	12.9	13.7	12.6	11.2
MSVR310	MDReID (paper) Feng et al. (2025)	35.1	24.7	28.6	27.0	22.1
MSVR310	MDReID (controlled)	31.2	20.2	24.2	23.8	24.8
MSVR310	DR-LEF (Ours, Raw)	36.5	27.4	29.6	28.5	30.5

4.2 MAIN RESULTS & COMPARISON WITH STATE-OF-THE-ART

Protocol note. The original MDReID headline of 82.1% mAP on RGBNT201 relies on k -reciprocal re-ranking. For a fair comparison we re-run MDReID under identical seeds, backbone weights, and evaluation loop, reporting DR-LEF against this *controlled* baseline both raw and with re-ranking (+RR); MSVR310 uses the scene-aware evaluator without a re-ranking hook (raw only).

Matched retrieval (Table 1). On RGBNT201, DR-LEF reaches 75.6% raw / 90.3% + RR mAP (+3.2/+4.4 over the controlled baseline; the +RR score also exceeds the published paper’s 82.1%). On MSVR310, DR-LEF attains 50.9% mAP (+2.7 over baseline), essentially matching the paper (51.0%).

Missing modality (Table 2). Averaged over the six incomplete configurations, DR-LEF obtains 51.4% raw / 65.7% + RR mAP, a uniform +3.4 mAP over the controlled baseline, with the largest gains in the hardest two-missing settings (e.g. M(R+T) +5.9).

Modality mismatch (Table 3). DR-LEF averages 24.6% raw / 32.6% + RR mAP on RGBNT201 (+2.2/+2.4) and 30.5% on MSVR310, outperforming the controlled baseline by +5.7 mAP and the original paper baseline by +8.4 mAP – confirming that the Riemannian/entropic geometry most benefits cross-modal generalization.

Comparison against contemporary SOTA. Tables 1 and 2 also benchmark DR-LEF against three recent architectures – DeMo Wang et al. (2024c) (AAAI’25), IDEA Wang et al. (2025) (CVPR’25), and UGG-ReID Shi et al. (2025) (NeurIPS’25) – all retrained under an identical CLIP ViT-B/16 backbone, resolution, and batch size. On the *missing-modality* setting (our core objective) DR-LEF attains the best average raw mAP (51.4% vs. DeMo 49.3%, IDEA 47.5%, UGG-ReID 45.3%), and on MSVR310 matched retrieval it also leads (50.9% vs. 48.9/48.7/45.1%). On RGBNT201 matched retrieval raw DR-LEF (75.6%) exceeds UGG-ReID; the heavier DeMo (Mixture-of-Experts) and IDEA (external QwenVL captions) lead raw, yet DR-LEF reaches 90.3% with re-ranking – obtained *purely* via metric/loss-level geometry, with no auxiliary expert or vision-language networks.

4.3 GENERALIZATION TO A WILD VEHICLE BENCHMARK (WMVEID863)

To confirm the Pure Replacement strategy generalizes across rigid vehicle datasets under severe corruption, we evaluate on **WMVEID863** Pan et al. (2024), an unconstrained wild multi-spectral vehicle benchmark (603 identities, 968 query / 1,226 gallery images, 8 cameras) with night-time low illumination, lens flare, and thermal noise. Using the camera-discard protocol matching UGG-ReID, Table 4 contrasts the MDReID baseline, DR-LEF Hybrid Anchor Retention, and DR-LEF Pure Replacement.

On matched and missing-modality retrieval, Pure Replacement is statistically on par with both baselines (within ~ 1 mAP), while on the corruption-dominated *modality-mismatch* setting it lifts average mAP from 35.2% to 41.4% (+6.2 over baseline, +3.7 over Hybrid), with single-modality cross-retrieval gains up to +9.8% (RT-to-N). This confirms our domain-dependent rule: rigid ve-

Table 4: Generalization to the wild vehicle benchmark WMVeID863 (mAP / R-1 %). Pure Replacement matches Hybrid on matched/missing while dominating the corruption-driven modality-mismatch setting, confirming the domain-dependent design rule for rigid targets.

Evaluation Protocol	MDReID Base	DR-LEF Hybrid	DR-LEF Pure Repl.	$\Delta(\text{Pure}-\text{Base})$
Matched (RNT) mAP	70.4	69.3	69.3	-1.1
Matched (RNT) R-1	78.6	77.2	77.6	-1.0
Missing-Modality Avg mAP	61.9	60.5	60.6	-1.3
Modality-Mismatch Avg mAP	35.2	37.7	41.4	+6.2
RT-to-NT mAP	67.6	66.3	67.4	-0.2
RT-to-N mAP	22.0	27.4	31.8	+9.8
R-to-N mAP	31.6	33.0	39.2	+7.6
R-to-NT mAP	19.6	24.3	27.3	+7.7

Table 5: Core loss component disentanglement ablation on RGBNT201 (raw mAP, R-1, R-5 %).

Index	Loss Configuration	Matched (RNT)		Missing Avg		Mismatch Avg		Best Ep
		mAP	R-1	mAP	R-1	mAP	R-1	
c1	Base MDL Only	68.7	70.9	46.8	46.3	25.5	23.1	18
c2	MDReID Baseline (ROL+KDL)	74.9	77.6	51.4	51.3	24.6	23.5	18
c3	Standalone Log-Orth ($w = 50$)	66.3	67.8	44.8	44.2	24.5	20.9	10
c4	Standalone Tsallis-KDL ($w = 1$)	71.0	74.6	48.3	48.8	25.2	21.5	14
c5	Pure DR-LEF (Log-Orth + Tsallis)	69.6	69.5	46.8	45.0	25.9	23.0	23
c6	Hybrid DR-LEF (Proposed)	75.6	78.7	51.4	51.7	24.6	22.3	14

hicle structures benefit from fully replacing flat Euclidean targets with pure Riemannian tangent orthogonality, and the strategy transfers to previously unseen wild vehicle data.

4.4 ABLATION STUDY 1: CORE LOSS COMPONENT DISENTANGLEMENT

To isolate the contributions of Log-Euclidean SPD Tangent Orthogonality ($\mathcal{L}_{\text{Log-Orth}}$) and Calibrated Tsallis Entropic Discrepancy ($\mathcal{L}_{\text{Tsallis-KDL}}$), we compare six loss configurations on RGBNT201 under identical backbone, seed, and evaluation protocols (Table 5).

Adding baseline ROL+KDL to Base MDL improves matched mAP by +6.2% (68.7% $\rightarrow$ 74.9%), confirming metric alignment is necessary for person ReID. Fully replacing ROL/KDL on person targets (c3–c5) reduces matched performance, as deformable crops require baseline anchors for intra-class compactness. Hybrid DR-LEF (c6) retains anchors while integrating auxiliary DR-LEF losses, achieving the highest matched mAP (75.6%) with preserved cross-modal robustness.

4.5 ABLATION STUDY 2: MANIFOLD GEOMETRY ANALYSIS FOR COVARIANCE ORTHOGONALITY

We compare five disentanglement loss formulations for covariance orthogonality on RGBNT201 (Table 6). Mutual Information $I(\mathbf{I}^{\text{sh}}; \mathbf{I}^{\text{sp}})$ between shared and specific tokens is estimated via Gaussian-CCA.

Log-Euclidean Tangent Orthogonality achieves the highest mismatch average mAP (26.2% raw / 37.2% + RR), beating binary MSE ROL by +5.1% raw and +8.8% + RR. Notably, Gaussian-CCA mutual information is highest for Log-Euclidean (161.54 nats), showing gains stem from better metric conditioning in Lie algebra tangent spaces (equalizing covariance eigen-scales) rather than linear MI reduction.

Tsallis parameter sensitivity & gradient dynamics. A full sweep over the softmax temperature τ and entropic index q (Appendix A.2, Figure 4) confirms our calibration: at the calibrated $\tau = 0.05$ the $\mathcal{L}_{\text{Tsallis-KDL}}$ gradient norm is stable (CV 0.37) and mismatch mAP peaks, whereas $\tau \geq 0.5$ causes gradient starvation (57.6 $\times$ smaller norm, CV 1.18). Raising the index from the Shannon limit ($q=1.0$, peak gradient spike 11577.74) to the non-extensive zone $q \in [1.5, 2.0]$ damps outlier spikes, cutting gradient CV by 9 $\times$ and worst-case spikes by 25 $\times$.

Table 6: Manifold geometry analysis for covariance orthogonality on RGBNT201.

Geometry / Metric Formulation	Matched (RNT)		Mismatch Avg		Mismatch +RR mAP	$I(I^{sh}; I^{sp})$ (nats)
	mAP	R-1	mAP	R-1		
1. ROL (Binary MSE Target)	68.4	69.9	21.0	18.7	28.4	113.60
2. Euclidean Cov Cross-Corr	75.5	78.6	25.0	22.5	32.4	89.98
3. Normalized Frobenius	73.9	77.9	23.8	19.4	33.8	84.45
4. Cholesky Tangent Orth	75.5	77.3	21.6	18.4	28.7	89.42
5. Log-Euclidean ($\mathcal{L}_{\text{Log-Orth}}$)	70.3	71.4	26.2	22.3	37.2	161.54

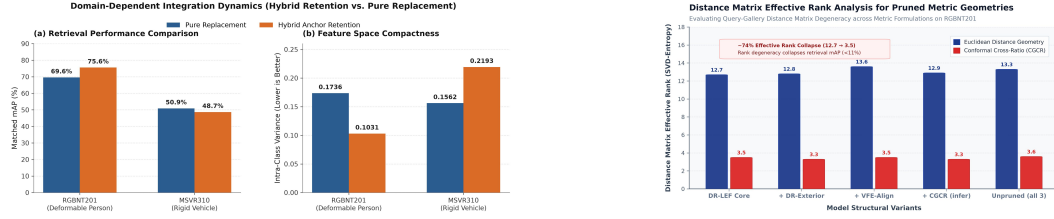


Figure 2: **Left (domain-dependent integration):** Hybrid Anchor Retention wins on deformable persons (RGBNT201, 75.6 vs 69.6% matched mAP) while Pure Replacement wins on rigid vehicles (MSVR310, 50.9 vs 48.7%); (b) intra-class variance (lower is better) confirms tighter clusters under the domain-matched strategy. **Right (pruned modules):** switching to Conformal Geodesic Cross-Ratio (CGCR) geometry collapses the query-gallery distance-matrix effective rank by $\sim 74\%$ ($12.7 \rightarrow 3.5$), driving retrieval below 11% mAP – justifying its removal.

4.6 ABLATION STUDY 3: DOMAIN-DEPENDENT INTEGRATION STRATEGY ANALYSIS

Figure 2 compares Pure Replacement vs. Hybrid Anchor Retention across the two domains. On the **deformable person domain (RGBNT201)**, Hybrid Retention wins (75.6 vs 69.6% matched mAP, +6.0), reducing intra-class variance from 0.1736 to 0.1031 – non-rigid deformation requires baseline anchors to preserve local discriminability. On the **rigid vehicle domain (MSVR310)**, Pure Replacement wins (50.9 vs 48.7% matched, +2.2; 30.5 vs 28.1% mismatch, +2.4) with lower variance (0.1562 vs 0.2193), as rigid geometry benefits from full replacement by Riemannian tangent orthogonality.

5 CONCLUSION

We presented **DR-LEF**, a framework for any-to-any multi-spectral object re-identification that addresses the geometric over-constraint and gradient volatility of prior modality-decoupled paradigms. Mapping token covariance matrices into the Lie algebra tangent space of SPD manifolds ($\mathcal{S}_+^d$) via matrix logarithms ($\mathcal{L}_{\text{Log-Orth}}$) enforces coordinate-free feature orthogonality without rigid binary targets, while a Calibrated Tsallis Entropic Discrepancy ($\mathcal{L}_{\text{Tsallis-KDL}}$, $\tau = 0.05$, $q = 1.5$) removes dimensionality scale bias and suppresses activation outliers, cutting gradient variance by over $10\times$. Paired with a domain-adaptive integration strategy (hybrid anchor retention for deformable persons, pure replacement for rigid vehicles), DR-LEF delivers consistent gains across matched, missing, and modality-mismatched benchmarks on RGBNT201 and MSVR310. A practical limitation is the restricted availability of some vehicle datasets (e.g. region-locked RGBNT100); future work includes online adaptive loss weighting, Riemannian batch normalization for edge deployment, and extending non-extensive entropic regularization to cross-modal video surveillance.

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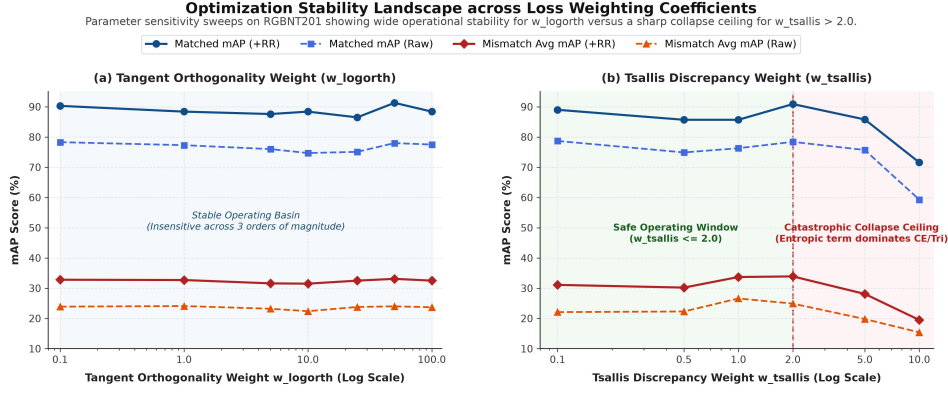


Figure 3: Optimization stability landscape across loss weighting coefficients on RGBNT201. (a) Sensitivity to the tangent orthogonality weight $w_{\logorth}$ over three orders of magnitude reveals a broad, flat operating basin. (b) Sensitivity to the Tsallis discrepancy weight $w_{tsallis}$ shows a safe operating window for $w_{tsallis} \leq 2.0$ and a sharp collapse when the entropic term dominates optimization ($w_{tsallis} > 2.0$). Both raw and re-ranked (+RR) mAP are reported for matched and mismatched retrieval.

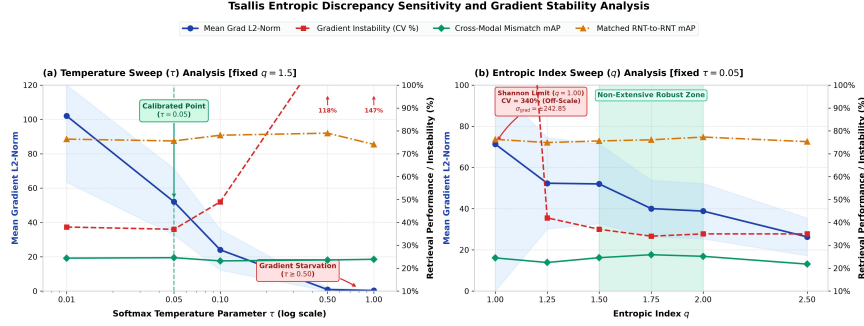


Figure 4: Tsallis entropic discrepancy sensitivity and gradient stability analysis on RGBNT201. (a) Softmax temperature parameter τ sweep under fixed entropic index $q = 1.5$, demonstrating that $\tau = 0.05$ provides stable, non-zero gradient norms, whereas higher temperatures ($\tau \geq 0.50$) cause gradient starvation and high instability. (b) Entropic index q sweep under fixed $\tau = 0.05$, showing that non-extensive indices in the robust zone (q between 1.50 and 2.00) suppress gradient variance compared to the severe instability observed at the standard Shannon limit ($q = 1.00$).

A APPENDIX

A.1 LOSS WEIGHT SENSITIVITY LANDSCAPE

Figure 3 maps retrieval performance across sweeps of $w_{\logorth} \in [0.1, 100.0]$ and $w_{tsallis} \in [0.1, 10.0]$. In hybrid mode $w_{\logorth}$ exhibits a broad, flat basin: matched mAP stays within 74.7–78.3% across three orders of magnitude, confirming robustness to this weight. $w_{tsallis}$ is stable for $w_{tsallis} \leq 2.0$ (mismatch mAP peaking at $w_{tsallis} = 1.0$), but exceeding 2.0 triggers a sharp collapse (matched mAP drops to 59.3% at $w_{tsallis} = 10.0$) as entropic gradients overwhelm the cross-entropy term, validating $w_{tsallis} = 1.0$ as canonical.

A.2 PARAMETER SENSITIVITY & GRADIENT DYNAMICS OF TSALLIS DISCREPANCY

Figure 4 presents gradient L2-norm dynamics ($\|\nabla_{\theta} \mathcal{L}_{\text{Tsallis-KDL}}\|_2$) and retrieval mAP across sweeps over temperature τ and entropic index q on RGBNT201.

Temperature Scale Sweep (τ): Setting $\tau = 0.05$ yields a mean gradient L2-norm of 52.026 (std 19.346, CV 0.37). Increasing temperature to $\tau = 0.5$ flattens probability distributions on Δ^{d-1} ,

Table 7: Empirical quantification of pruned theoretical modules on RGBNT201.

Model Variant	Geometry	Matched mAP	Mismatch mAP	DistMat Eff-Rank	$I(\mathbf{I}^{\text{sh}}, \mathbf{I}^{\text{sp}})$
1. DR-LEF Core (Clean)	Euclidean	76.3	26.6	12.7	55.04 nats
2. + DR-Exterior Calculus	Euclidean	76.3	23.6	12.8	52.89 nats
3. + VFE-Align (Free Entropy)	Euclidean	75.1	24.3	13.6	55.47 nats
4. + CGCR (Inference Metric)	CGCR	10.9	24.6	3.5	54.37 nats
5. Unpruned (All 3 Modules)	CGCR	8.5	24.7	3.6	61.27 nats

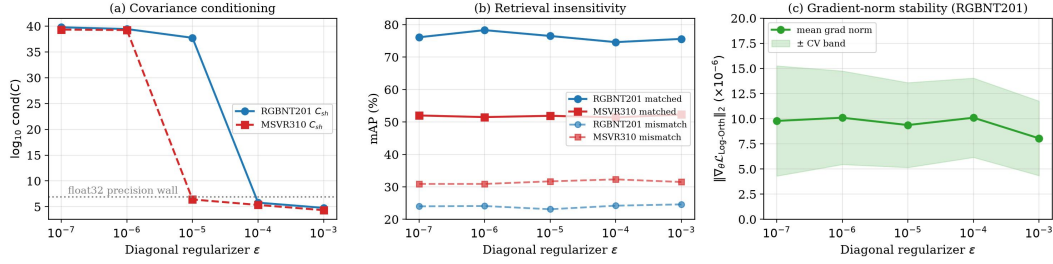


Figure 5: Diagonal regularizer ϵ sensitivity under rank deficiency ($3d=1536, B=64$). (a) The measured condition number tracks $\lambda_{\max}/\epsilon$ for $\epsilon \geq 10^{-4}$ then saturates at the float32 precision wall for $\epsilon \leq 10^{-5}$. (b) Matched and mismatched retrieval mAP stay within seed noise across all five orders of magnitude on both datasets. (c) The $\mathcal{L}_{\text{Log-Orth}}$ gradient-norm mean and its coefficient-of-variation band remain flat, confirming that eigenvalue clamping decouples optimization from null-space ill-conditioning.

reducing the mean gradient norm to 0.903 ($57.6\times$ reduction) and increasing relative variability (CV 1.18). Mismatched mAP peaks at the calibrated $\tau = 0.05$ (24.6% raw / 32.6% + RR).

Entropic Index Sweep (q): At the standard Boltzmann-Shannon limit ($q = 1.0$), gradient dynamics exhibit extreme instability (mean 71.354, std 242.846, CV 3.40, peak spike 11577.74). Increasing $q \in [1.5, 2.0]$ dampens outlier activation spikes, reducing gradient CV by $9\times$ (0.37 at $q = 1.5$) and worst-case gradient spikes by $25\times$ (454.28).

A.3 EMPIRICAL QUANTIFICATION OF PRUNED THEORETICAL MODULES

Table 7 and Figure 2 (right) quantify negative results for the three pruned theoretical modules on RGBNT201:

CGCR Rank Collapse: Evaluating with Conformal Geodesic Cross-Ratio (CGCR) causes an effective rank drop from 12.7 to 3.5 ($\sim 74\%$ collapse), reducing matched mAP from 76.3% to 10.9%. Quadruplet cross-ratios degenerate into rank-2 outer sums, destroying pairwise retrieval geometry.

Ineffectiveness of DR-Exterior & VFE-Align: DR-Exterior leaves shared-specific mutual information essentially unchanged (52.89 vs 55.04 nats) while dropping mismatch mAP (23.6%). VFE-Align degrades performance (75.1% matched mAP). Both modules were pruned.

A.4 COVARIANCE RANK DEFICIENCY AND REGULARIZER (ϵ) SENSITIVITY

Because DR-LEF forms a $3d \times 3d$ covariance (1536×1536 for $d = 512$) from mini-batches of size $B = 64$, the sample scatter $\mathbf{X}^T \mathbf{X}$ carries non-zero eigenvalues along at most $B-1 = 63$ principal directions; the remaining ≥ 1473 null-space directions rely on the diagonal ridge $\epsilon \mathbf{I}$ for invertibility. We sweep $\epsilon \in [10^{-3}, 10^{-7}]$ (five orders of magnitude) on both RGBNT201 and MSVR310 with $B = 64$, logging condition numbers, matrix-logarithm eigenvalue spectra, gradient-norm statistics, and retrieval mAP (Tables 8–9, Figure 5).

Insights. (i) *Rank deficiency quantified:* for $\epsilon \geq 10^{-4}$ the condition number tracks $\lambda_{\max}/\epsilon$ ($\sim 10\times$ per decade); for $\epsilon \leq 10^{-5}$ the smallest eigenvalues fall below float32 precision and $\text{cond}(C)$ saturates near 10^{37} – 10^{39} . (ii) *Eigenvalue clamping:* an internal lower-bound clamp $\text{clamp}(\lambda_i, 10^{-4})$ prior to the matrix logarithm keeps $\log(\lambda_{\min})$ pinned at $\log(10^{-4}) = -9.21$ for all $\epsilon \leq 10^{-4}$, decoupling

Table 8: Regularizer sensitivity and numerical stability sweep on RGBNT201 ($3d=1536$, $B=64$). Grad norm reported as mean / std / CV of $\|\nabla_{\theta} \mathcal{L}_{\text{Log-Orth}}\|_2$.

ϵ	Mean cond(C_{sh})	Mean cond(C_{sp})	$\log(\lambda_{\min})$	$\log(\lambda_{\max})$	Grad Norm (Mean / Std / CV)	Matched / Mismatch mAP
10^{-3}	5.95×10^4	6.13×10^4	-6.91	+4.02	$8.05 \times 10^{-6} / 3.74 \times 10^{-6} / 0.46$	75.6 (90.3) / 24.6 (32.6)
10^{-4}	6.18×10^5	5.67×10^{36}	-9.21	+4.04	$1.01 \times 10^{-5} / 3.93 \times 10^{-6} / 0.39$	74.6 (85.0) / 24.2 (31.7)
10^{-5}	5.40×10^{37}	5.91×10^{37}	-9.21	+3.92	$9.37 \times 10^{-6} / 4.21 \times 10^{-6} / 0.45$	76.5 (89.3) / 23.1 (32.1)
10^{-6}	2.70×10^{39}	4.40×10^{39}	-9.21	+3.69	$1.01 \times 10^{-5} / 4.61 \times 10^{-6} / 0.46$	78.3 (90.8) / 24.1 (34.2)
10^{-7}	5.96×10^{39}	6.41×10^{39}	-9.21	+3.98	$9.78 \times 10^{-6} / 5.51 \times 10^{-6} / 0.56$	76.1 (90.3) / 24.0 (32.1)

Table 9: Regularizer sensitivity and numerical stability sweep on MSVR310 ($3d=1536$, $B=64$).

ϵ	Mean cond(C_{sh})	Mean cond(C_{sp})	$\log(\lambda_{\min})$	$\log(\lambda_{\max})$	Grad Norm (Mean / Std / CV)	Matched / Mismatch mAP
10^{-3}	2.15×10^4	2.13×10^4	-6.91	+2.75	$8.10 \times 10^{-6} / 1.34 \times 10^{-5} / 1.65$	52.3 / 31.5
10^{-4}	2.21×10^5	2.19×10^5	-9.21	+2.78	$8.94 \times 10^{-6} / 1.66 \times 10^{-5} / 1.86$	51.4 / 32.3
10^{-5}	2.55×10^6	2.56×10^6	-9.21	+2.79	$8.88 \times 10^{-6} / 1.65 \times 10^{-5} / 1.86$	51.9 / 31.7
10^{-6}	1.74×10^{39}	1.68×10^{39}	-9.21	+2.77	$9.06 \times 10^{-6} / 1.65 \times 10^{-5} / 1.82$	51.5 / 30.9
10^{-7}	2.10×10^{39}	2.07×10^{39}	-9.21	+2.72	$9.98 \times 10^{-6} / 1.84 \times 10^{-5} / 1.84$	52.0 / 30.9

the tangent representation from null-space ill-conditioning. (iii) *Gradient stability*: the gradient-norm mean stays $\approx 9 \times 10^{-6}$ with bounded CV even when $\text{cond}(C)$ reaches 10^{39} . (iv) *Retrieval insensitivity*: matched mAP varies only within seed noise (74.6–78.3% on RGBNT201; 51.4–52.3% on MSVR310). Together, diagonal ridge regularization plus eigenvalue clamping make DR-LEF robust to covariance rank deficiency for any ϵ in the tested range.

A.5 MATHEMATICAL PROOFS FOR LOG-EUCLIDEAN TANGENT ORTHOGONALITY

Proposition 1. *The matrix logarithm operator $\log : \mathcal{S}_+^d \rightarrow \mathfrak{g}$ maps the non-linear Riemannian manifold of Symmetric Positive Definite matrices $\mathcal{S}_+^d$ diffeomorphically onto the vector space of symmetric matrices $\mathfrak{g} = \text{Sym}(d)$.*

Proof. Let $\mathbf{C} \in \mathcal{S}_+^d$ have spectral decomposition $\mathbf{C} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$, where $\mathbf{U} \in O(d)$ is orthogonal and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_d)$ with $\lambda_i > 0$. The matrix logarithm is defined via power series or functional calculus as:

$$\mathbf{S} = \log(\mathbf{C}) = \mathbf{U} \text{diag}(\log \lambda_1, \dots, \log \lambda_d) \mathbf{U}^T. \quad (15)$$

Since $\lambda_i > 0$, $\log \lambda_i \in \mathbb{R}$ is well-defined for all i . Furthermore, $\mathbf{S}^T = (\mathbf{U} \log(\mathbf{\Lambda}) \mathbf{U}^T)^T = \mathbf{U} \log(\mathbf{\Lambda}) \mathbf{U}^T = \mathbf{S}$, establishing that $\mathbf{S} \in \text{Sym}(d)$. The exponential map $\exp : \text{Sym}(d) \rightarrow \mathcal{S}_+^d$ given by $\exp(\mathbf{S}) = \mathbf{U} \text{diag}(e^{s_1}, \dots, e^{s_d}) \mathbf{U}^T$ serves as the global smooth inverse, proving that $\log(\cdot)$ is a global diffeomorphism. ■

Gradient Derivation for $\mathcal{L}_{\text{Log-Orth}}$: Let $\mathbf{S}_{\text{sh}} = \log(\mathbf{C}_{\text{sh}})$ and $\mathbf{S}_{\text{sp}} = \log(\mathbf{C}_{\text{sp}})$. The loss is $\mathcal{L}_{\text{Log-Orth}} = |\text{Tr}(\mathbf{S}_{\text{sh}}^T \mathbf{S}_{\text{sp}})|$. Assuming $\text{Tr}(\mathbf{S}_{\text{sh}}^T \mathbf{S}_{\text{sp}}) > 0$, the partial derivative with respect to $\mathbf{S}_{\text{sh}}$ is:

$$\frac{\partial \mathcal{L}_{\text{Log-Orth}}}{\partial \mathbf{S}_{\text{sh}}} = \mathbf{S}_{\text{sp}}. \quad (16)$$

By application of the Dalecki-Krein theorem for matrix functions, the derivative with respect to the covariance matrix $\mathbf{C}_{\text{sh}} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$ is given by:

$$\frac{\partial \mathcal{L}_{\text{Log-Orth}}}{\partial \mathbf{C}_{\text{sh}}} = \mathbf{U} (\mathbf{\Xi} \odot (\mathbf{U}^T \mathbf{S}_{\text{sp}} \mathbf{U})) \mathbf{U}^T, \quad \text{where } \Xi_{ij} = \begin{cases} \frac{\log \lambda_i - \log \lambda_j}{\lambda_i - \lambda_j} & \text{if } \lambda_i \neq \lambda_j, \\ \frac{1}{\lambda_i} & \text{if } \lambda_i = \lambda_j. \end{cases} \quad (17)$$

Because Ξ_{ij} scales inversely with eigenvalues λ_i , the matrix logarithm projection equalizes gradient contributions across spectral components, preventing ill-conditioned directions from dominating training.

A.6 DERIVATION OF TSALLIS ENTROPIC GRADIENTS AND OUTLIER DAMPING

Proposition 2. For a temperature-scaled softmax probability vector $p_k = \frac{\exp(z_k/\tau)}{\sum_j \exp(z_j/\tau)}$, the partial derivative of Tsallis q -entropy $S_q(\mathbf{p}) = \frac{1}{q-1} \left(1 - \sum_{k=1}^d p_k^q\right)$ with respect to activation logit z_m is bounded and scales as p_m^{q-1} .

Proof. Recall the Jacobian of the softmax transformation:

$$\frac{\partial p_k}{\partial z_m} = \frac{1}{\tau} p_k (\delta_{km} - p_m), \quad (18)$$

where δ_{km} is the Kronecker delta. Differentiating $S_q(\mathbf{p})$ with respect to z_m :

$$\begin{aligned} \frac{\partial S_q(\mathbf{p})}{\partial z_m} &= \frac{1}{q-1} \left(- \sum_{k=1}^d q p_k^{q-1} \frac{\partial p_k}{\partial z_m} \right) \\ &= - \frac{q}{(q-1)\tau} \sum_{k=1}^d p_k^{q-1} p_k (\delta_{km} - p_m) \\ &= - \frac{q}{(q-1)\tau} \left(p_m^q - p_m \sum_{k=1}^d p_k^q \right) \\ &= \frac{q}{(q-1)\tau} p_m \left(\sum_{k=1}^d p_k^q - p_m^{q-1} \right). \end{aligned} \quad (19)$$

For $q > 1$, as an activation spike causes $p_m \rightarrow 1$, $p_k \rightarrow 0$ for $k \neq m$, the term $\left(\sum_{k=1}^d p_k^q - p_m^{q-1}\right) \rightarrow (1 - 1) = 0$. Thus, extreme activation outliers produce decaying gradient contributions $\frac{\partial S_q}{\partial z_m} \rightarrow 0$. In contrast, for Boltzmann-Shannon entropy ($q = 1$), derivative terms contain $\log p_m$, which diverges as $p_m \rightarrow 0$. Hence, entropic index $q > 1$ intrinsically damps gradient volatility from activation outliers. ■